

AYMANE AIT DADS

Data Science Intern | AI Evaluation Systems · Machine Learning · Python & SQL Analytics

Ayman.Ait-dads@eurecom.fr | +33 7 60 92 50 93 | linkedin.com/in/aymane-ait-dads | Sophia Antipolis, France

PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams (score 0.86/1.0, \$106K+ prize pool) in the NVIDIA Nemotron Reasoning Challenge by building end-to-end LLM fine-tuning and evaluation pipelines on a 30B-parameter model - directly demonstrating the ability to measure AI quality and identify failure modes at scale. Automated network quality analysis at Orange Maroc across 100,000+ records using Random Forest and K-Means models, reducing manual analysis time by 60% and delivering actionable maintenance recommendations to stakeholders. Coursework and project work at EURECOM spans machine learning, statistical modeling, NLP, and data mining - with hands-on experience in Python, SQL, ensemble methods, and classical ML techniques aligned with the Adobe Data Science team's toolkit. Aims to contribute to Adobe Experience Platform's AI Assistant evaluation frameworks by applying rigorous hypothesis-driven analysis and automated pipeline design to identify user interaction patterns and AI performance failure modes.

CORE COMPETENCIES

AI quality measurement | evaluation frameworks | machine learning models | user interaction data analysis | LLM-as-a-judge systems | statistical modeling | random forests | failure mode identification

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built **Random Forest and K-Means models** on 100,000+ fiber-optic network records - classifying network quality across thousands of nodes using upload speed, latency, and jitter features.
- Developed a clustering pipeline that mapped customer performance profiles to **5 commercial service tiers**, with output consumed directly by the network optimization team for infrastructure decisions.
- Reduced manual data analysis time by **~60%** through automated feature engineering, model evaluation workflows, and reusable Pandas-based preprocessing pipelines in Python.
- Delivered stakeholder presentations translating model outputs into **maintenance recommendations**, connecting statistical modeling results to actionable go-to-market and operational strategies.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge

Kaggle Competition · In Progress (June 2026) · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) in the NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT ($r=32$, BF16, completion-only loss, 8,192-token context).
- Ran **controlled ablation experiments** across LR schedules, packing strategies, and synthetic augmentation - discovering that reasoning-style consistency outweighs data volume in LLM SFT; built custom cipher parsers and 8-bit operation solvers to generate verified chain-of-thought evaluation data.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Kaggle GPU

AI-Generated Image Detection

EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in Class

- Ranked **1st in EURECOM class** private Kaggle leaderboard (0.791 AUC) by fine-tuning a CLIP ViT-L/14 backbone (428M parameters) with a custom MLP head trained on 250,000 images across 25+ generator types.
- Discovered that **1-epoch training outperformed 4-epoch training** (0.793 vs. 0.767 AUC) - a key finding on out-of-distribution generalization; submitted independently to the NTIRE 2026 @ CVPR CodaBench international benchmark.

PyTorch · OpenCLIP · FP16 mixed precision · AdamW · Kaggle

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Statistics, NLP, Computer Vision, Cloud Computing, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

ML Frameworks & Data Science: Python | SQL | Scikit-learn | PyTorch | Pandas | NumPy | Random Forests | Ensemble Methods

AI Evaluation & LLM Systems: LoRA SFT | Hugging Face Transformers | TRL | PEFT | vLLM | Prompt Engineering | Chain-of-Thought

Analytics, MLOps & Visualization: Feature Engineering | Ablation Studies | Power BI | Matplotlib | Git | Linux | Kaggle GPU